

Binary Outcome Models

ECMT 676 (Econometric II) Lecture Notes

TAMU

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- The most common application of binary response models is when we are interested in “explaining” a binary outcome in terms of some explanatory variables. Thus, we are interested in a conditional probability.
- A less common application is when we have a linear model of an underlying quantitative variable, but the data collection scheme censors the data.
- For example, we have a linear model for willingness to pay for a project or product.
- However, because it is difficult to elicit WTP, each individual may be presented with a cost of the project; we then only observe whether they are in favor of the project at that cost.

- We treat data censoring problems later.
- For now, we focus on the first situation. So, y is a binary (zero-one) variable.
- For example, $y = \textit{employed}$ or $y = \textit{arrested}$.
- Given a set of (exogenous) covariates $\mathbf{x}$, we are interested in

$$P(y = 1|\mathbf{x}) = p(\mathbf{x}),$$

which is called the *response probability*. It is the probability of a “success,” that is, $y = 1$.

- As in regression, we are interested in the partial effects of the x_j on $p(\mathbf{x})$. For continuous x_j , these are usually

$$\frac{\partial p(\mathbf{x})}{\partial x_j}.$$

- For discrete x_j , look at changes in the response probability (usually holding other variables fixed).
- For example, if $x_K = \text{train}$ (job training indicator) and y is an employment indicator,

$$p(x_1, \dots, x_{K-1}, 1) - p(x_1, \dots, x_{K-1}, 0)$$

is the effect of job training on the employment probability, at given values for the other covariates.

- In nonlinear models generally, and binary response models specifically, it is often useful to have a single number to summarize the relationship between $P(y = 1|\mathbf{x})$ and x_j . *In a linear model that is simply the coefficient.*

- Generally, we might report an estimated *average partial effect (APE)*.
- The APE for a continuous x_j is

$$E_{\mathbf{x}} \left[\frac{\partial p(\mathbf{x})}{\partial x_j} \right],$$

which means we average the partial effect across the population distribution of $\mathbf{x}$. This is a weighted average of the partial effects at each outcome $\mathbf{x}$.

- Suppose x_K is a binary variable. Then its APE is

$$E_{\mathbf{x}_{(K)}} [p(\mathbf{x}_{(K)}, 1) - p(\mathbf{x}_{(K)}, 0)]$$

where $\mathbf{x}_{(K)}$ is the $1 \times K$ vector with x_K excluded.

- Another partial effect that has been reported in empirical work is the *partial effect at the average (PEA)*.
- For a continuous variable x_j ,

$$\frac{\partial p(\mu_{\mathbf{x}})}{\partial x_j}.$$

- In nonlinear models, the APE and PEA can be very different: the expected value does not pass through nonlinear functions.
- Because $\mu_{\mathbf{x}}$ might not even represent a population unit – for example, if $\mathbf{x}$ includes discrete variables, such as dummy variables – the PEA might not be especially interesting.

- Some simple, useful facts about Bernoulli (zero-one) random variables are

$$\begin{aligned}E(y|\mathbf{x}) &= P(y = 1|\mathbf{x}) = p(\mathbf{x}) \\ \text{Var}(y|\mathbf{x}) &= p(\mathbf{x})[1 - p(\mathbf{x})]\end{aligned}$$

- So a binary variable has natural heteroskedasticity except in the special case where $p(\mathbf{x})$ does not depend on $\mathbf{x}$.
- Unlike variables that take on more than two values, there is a necessary link between the mean and the variance.
- It is not possible for $E(y|\mathbf{x}) = p(\mathbf{x})$ while $\text{Var}(y|\mathbf{x}) \neq p(\mathbf{x})[1 - p(\mathbf{x})]$.

Linear Probability Model (LPM)

- LPM models the response probability as a function linear in parameters.
- Absorbing an intercept into $\mathbf{x}$, if we take the model literally we are assuming

$$P(y = 1|\mathbf{x}) = \beta_1 + \beta_2 x_2 + \dots + \beta_K x_K \equiv \mathbf{x}\boldsymbol{\beta}.$$

- Because this is also $E(y|\mathbf{x})$, we can use OLS to consistently estimate $\boldsymbol{\beta}$. In fact, if the conditional mean is truly $\mathbf{x}\boldsymbol{\beta}$, the OLS estimator is unbiased.

- Because $Var(y|\mathbf{x}) = \mathbf{x}\beta(1 - \mathbf{x}\beta)$ – a rare case where we know the functional form of heteroskedasticity – inference for OLS should be made robust to heteroskedasticity. As we know, this is easy to do.
- Because y is binary, we must rely on large-sample properties for inference; clearly normality of $D(y|\mathbf{x})$ does not hold.
- The LPM is always a good starting point when y is the variable we hope to explain. The estimated coefficients give direct estimates of the effects of each x_j on the response probability. (Of course, as with any regression framework, we can include various functional forms in $\mathbf{x}$, such as quadratics, interactions, and dummy variables.)

- The LPM is simple to estimate and interpret. The often cited drawbacks of the LPM include

(1) Nothing guarantees the OLS fitted values, $\hat{y}_i = \mathbf{x}_i\hat{\boldsymbol{\beta}}$, are in the unit interval. As these are estimates of the $p(\mathbf{x}_i)$, one might worry about estimated probabilities above one or negative.

(2) While we can use various functional forms in $\mathbf{x}$, it is difficult to impose diminishing effects of the x_j on the $p(\mathbf{x})$. For example, if $\beta_j > 0$, increasing x_j increases $p(\mathbf{x}) = \mathbf{x}\boldsymbol{\beta}$ by β_j , no matter the values of x_j or the other elements of $\mathbf{x}$. Logically, the effect must diminish at some point.

(3) Heteroskedasticity. This has asymptotic efficiency implications *if* we assume that $p(\mathbf{x}) = \mathbf{x}\boldsymbol{\beta}$. That is, in principle we can improve efficiency by weighted least squares, but $\mathbf{x}_i\hat{\boldsymbol{\beta}}$ not strict between zero and one for all i causes problems because the efficient weights are supposed to be $1/[\mathbf{x}_i\hat{\boldsymbol{\beta}}(1 - \mathbf{x}_i\hat{\boldsymbol{\beta}})]$.

- As a practical matter, it makes more sense to think of the LPM as the best linear approximation (in a mean squared error sense) to the true response probability, $p(\mathbf{x})$.
- That is,

$$\begin{aligned} y &= \mathbf{x}\beta + u \\ E(\mathbf{x}'u) &= \mathbf{0} \end{aligned}$$

is all we are willing to assume.

- If so, then $E(u^2|\mathbf{x})$ generally depends on $p(\mathbf{x})$ in addition to $\mathbf{x}\beta$, but the heteroskedasticity-robust variance matrix estimator is still valid (because it is valid for heteroskedasticity of unknown form).

- A carefully chosen linear model can yield good estimates of the APEs defined earlier. In other words, the LPM often yields good estimates of *average* effects.
- A leading reason for going from the LPM to nonlinear models of $p(\mathbf{x})$ is to allow the partial effects to vary across different values of $\mathbf{x}$.

Index Models: Probit and Logit

- A general index model has the form

$$P(y = 1|\mathbf{x}) = G(\mathbf{x}\boldsymbol{\beta})$$

for some $G : \mathbb{R} \rightarrow (0, 1)$.

- In most cases, $G(\cdot)$ is actually a CDF for a continuous random variable with density $g(\cdot)$. Then, $G(\cdot)$ is strictly increasing, and the estimates are easier to interpret.
- The leading cases are $G(z) = \Phi(z)$ (probit) and $G(z) = \exp(z)/[1 + \exp(z)]$ (logit).

- MLE is straightforward. The general log likelihood for random draw i is

$$\ell_i(\boldsymbol{\beta}) = (1 - y_i) \log[1 - G(\mathbf{x}_i\boldsymbol{\beta})] + y_i \log[G(\mathbf{x}_i\boldsymbol{\beta})].$$

- Asymptotic variance has the same form as for probit:

$$\left(\sum_{i=1}^N \frac{[g(\mathbf{x}_i\hat{\boldsymbol{\beta}})]^2 \mathbf{x}_i' \mathbf{x}_i}{G(\mathbf{x}_i\hat{\boldsymbol{\beta}})[1 - G(\mathbf{x}_i\hat{\boldsymbol{\beta}})]} \right)^{-1},$$

where

$$g(z) = \phi(z) \text{ for probit}$$

$$g(z) = \exp(z) / [1 + \exp(z)]^2 \text{ for logit}$$

- Testing multiple hypotheses about β – usually joint exclusion restrictions – is most easily done with the Wald and LR statistics.
- The score statistics are convenient for testing the standard index models against more complicated alternatives (below).

- More interesting is: What do we do with the β estimates?
- Let x_j be continuous. Then

$$\frac{\partial p(\mathbf{x})}{\partial x_j} = \beta_j g(\mathbf{x}\beta).$$

- because $g(z) > 0$ (assume it is a continuous density), β_j gives the direction of the partial effect. But its magnitude depends on $g(\mathbf{x}\beta)$.
- For probit, the largest value of the scale factor is about $.4 = g(0)$. For logit, it is $.25$.

- For two continuous covariates, the ratio of the coefficients give the ratio of the partial effects, independent of $\mathbf{x}$.

$$\frac{\partial p(\mathbf{x})/\partial x_j}{\partial p(\mathbf{x})/\partial x_h} = \frac{\beta_j g(\mathbf{x}\beta)}{\beta_h g(\mathbf{x}\beta)} = \beta_j/\beta_h.$$

- No simple relationship exists for discrete variables or changes.
- In any case, we would like the magnitude of the effect.

- Two common summary measures are the estimated PEAs and APEs.
- The estimated PEA for a continuous variable is

$$\widehat{PEA}_j = \hat{\beta}_j g(\bar{x}\hat{\beta})$$

- As discussed earlier, putting in averages for discrete covariates might not be especially interesting.
- When $\mathbf{x}$ includes nonlinear functions, such as age^2 , probably makes more sense to use $(\overline{age})^2$ rather than average age_i^2 .
- Delta method or bootstrapping can be used to get a standard error for $\widehat{PEA}_j$.

- The APE has more appeal, as we are averaging partial effects for actual units:

$$\widehat{APE}_j = \hat{\beta}_j \left[N^{-1} \sum_{i=1}^N g(\mathbf{x}_i \hat{\beta}) \right]$$

- To use the delta method, must account for randomness in $\mathbf{x}_j$, too.

- Whether we use the PEA or APE, the scale factor multiplying $\hat{\beta}_j$ is below one, and sometimes well below one.
- It makes no sense to compare magnitudes of coefficients across probit, logit, LPM. Comparing APEs is preferred.
- In particular, if $\hat{\gamma}_j$ is the linear regression coefficient on x_j from estimating an LPM, it can be compared with $\widehat{APE}_j$ (provided no other function of x_j appears in the regressors).

- Suppose x_K is a binary variable. Then its APE is estimated as

$$\widehat{APE}_K = N^{-1} \sum_{i=1}^N [G(\mathbf{x}_{i(K)} \hat{\beta}_{(K)} + \hat{\beta}_K) - G(\mathbf{x}_{i(K)} \hat{\beta}_{(K)})],$$

where $\mathbf{x}_{i(K)}$ is $\mathbf{x}_i$ but without x_{iK} .

- The APE has a nice counterfactual interpretation that is especially useful in policy analysis. Called the *average treatment effect (ATE)* in the treatment effect literature with a binary outcome. (The “treatment,” x_K , is binary.)
- Can average the individual treatment effects across subgroups, too, or insert fixed values for some of the other covariates.

- For a discrete x_K , the estimated PEA is

$$\widehat{PEA}_K = G(\bar{x}_{(K)}\hat{\beta}_{(K)} + \hat{\beta}_K) - G(\bar{x}_{(K)}\hat{\beta}_{(K)})$$

Again, this might correspond to a weird population unit, or might not be representative of the population.

- Complicated functional forms are, in principle, easily handled within the index structure.
- For example, suppose

$$P(y = 1|\mathbf{z}) = G[\beta_0 + \beta_1 z_1 + \beta_2 z_1^2 + \beta_3 \log(z_2) + \beta_4 z_3] \equiv G(\mathbf{x}\boldsymbol{\beta})$$

- Then

$$\frac{\partial P(y = 1|\mathbf{z})}{\partial z_1} = (\beta_1 + 2\beta_2 z_1)g(\mathbf{x}\boldsymbol{\beta})$$

$$\frac{\partial P(y = 1|\mathbf{z})}{\partial z_2} = (\beta_3 / z_2)g(\mathbf{x}\boldsymbol{\beta})$$

$$\frac{\partial \log P(y = 1|\mathbf{z})}{\partial \log z_2} = \beta_3 g(\mathbf{x}\boldsymbol{\beta}) / G(\mathbf{x}\boldsymbol{\beta})$$

- The signs of the coefficients are informative, but the partial effects are somewhat complicated. Need to evaluate them at interesting values or average across the distribution of $\mathbf{x}$ similar to the usual APE calculation.
- For example, the average elasticity of $P(y = 1|\mathbf{z})$ with respect to z_2 is

$$\hat{\beta}_3 \left[N^{-1} \sum_{i=1}^N g(\mathbf{x}_i \hat{\beta}) / G(\mathbf{x}_i \hat{\beta}) \right].$$

Specification Issues and Testing

- There is much confusion about specification issues in probit, logit, and other models, because sometimes inappropriate parallels are made with linear models.
- Probit is easiest to discuss because analytical results are available. Qualitatively, the findings for probit carry over to other nonlinear binary response models.

Omitted Variable Independent of Covariates

- Consider first the problem of an omitted variable independent of $\mathbf{x}$, call it c :

$$\begin{aligned}P(y = 1|\mathbf{x}, c) &= \Phi(\mathbf{x}\boldsymbol{\beta} + c) \\c|\mathbf{x} &\sim \text{Normal}(0, \sigma_c^2)\end{aligned}$$

where $\mathbf{x}$ includes unity so $E(c) = 0$ is without loss of generality.

- Write the underlying latent variable as

$$y^* = \mathbf{x}\boldsymbol{\beta} + c + e, \quad D(e|\mathbf{x}, c) = \text{Normal}(0, 1)$$

- It follows that

$$D(\underbrace{c + e}_{=v}|\mathbf{x}) = \text{Normal}(0, \sigma_c^2 + 1).$$

- Therefore,

$$\begin{aligned}P(y = 1|\mathbf{x}) &= P(\mathbf{x}\boldsymbol{\beta} + c + e > 0|\mathbf{x}) \\&= P(v > -\mathbf{x}\boldsymbol{\beta}|\mathbf{x}) \\&= 1 - P(v < -\mathbf{x}\boldsymbol{\beta}|\mathbf{x}) \\&= 1 - P(v/(1 + \sigma_c^2)^{1/2} < -\mathbf{x}\boldsymbol{\beta}/(1 + \sigma_c^2)^{1/2}|\mathbf{x}) \\&= 1 - \Phi[-\mathbf{x}\boldsymbol{\beta}/(1 + \sigma_c^2)^{1/2}] \\&= \Phi[\mathbf{x}\boldsymbol{\beta}/(1 + \sigma_c^2)^{1/2}].\end{aligned}$$

- Define the scaled parameter vector

$$\beta_c \equiv \beta / (1 + \sigma_c^2)^{1/2},$$

so we can write $P(y = 1|\mathbf{x}) = \Phi(\mathbf{x}\beta_c)$.

- It follows immediately that probit of y_i on $\mathbf{x}_i$ consistently estimates β_c .
- That β_c is attenuated toward zero has been called “attenuation bias.”
- This “bias” would not happen in a linear model because $E(y|\mathbf{x}) = \mathbf{x}\beta + E(c|\mathbf{x}) = \mathbf{x}\beta$.

Heteroskedasticity in the Latent Variable Model

- Again suppose y is the variable of interest, and now we allow heteroskedasticity in the error e in

$$y = 1[\mathbf{x}\boldsymbol{\beta} + e > 0].$$

- Suppose we assume

$$e|\mathbf{x} \sim \text{Normal}(0, \exp(2\mathbf{x}_1\delta)),$$

where $\mathbf{x}_1$ is a subset of $\mathbf{x}$ (and does not include a constant).

- So homoskedasticity is $\delta = 0$ and then e has unit variance.

- The introduction of heteroskedasticity in e changes the response probability, $P(y = 1|\mathbf{x})$:

$$\begin{aligned} P(y = 1|\mathbf{x}) &= P(e > -\mathbf{x}\boldsymbol{\beta}|\mathbf{x}) = P[\exp(-\mathbf{x}_1\delta)e > -\exp(-\mathbf{x}_1\delta)\mathbf{x}\boldsymbol{\beta}|\mathbf{x}] \\ &= 1 - \Phi[-\exp(-\mathbf{x}_1\delta)\mathbf{x}\boldsymbol{\beta}] = \Phi[\exp(-\mathbf{x}_1\delta)\mathbf{x}\boldsymbol{\beta}]. \end{aligned}$$

- δ can be estimated along with $\boldsymbol{\beta}$ using the Bernoulli MLE, as before.
- The MLE asymptotic theory is standard.

- Now, the derivatives and changes in $P(y = 1|\mathbf{x})$ are much more complicated, and need not have the same sign as the relevant coefficient.
- For example, if x_K is in $\mathbf{x}_1$,

$$\begin{aligned}\frac{\partial P(y = 1|\mathbf{x})}{\partial x_K} &= \phi[\exp(-\mathbf{x}_1\delta)\mathbf{x}\beta] \{\beta_K \exp(-\mathbf{x}_1\delta) - \delta_K \exp(-\mathbf{x}_1\delta)\mathbf{x}\beta\} \\ &= \phi[\exp(-\mathbf{x}_1\delta)\mathbf{x}\beta] \exp(-\mathbf{x}_1\delta) \{\beta_K - \delta_K \mathbf{x}\beta\}\end{aligned}$$

- If we view $P(y = 1|\mathbf{x}) = \Phi[\exp(-\mathbf{x}_1\delta)\mathbf{x}\beta]$ as just a way to generalize functional form, partial effects should be computed.

- An alternative to generalizing functional form is to include covariates in a flexible way in probit and logit.
- After estimation of, say, probit with squares and interactions, it is legitimate to compare log likelihood with the heteroskedastic probit log likelihood.
- Generally, heteroskedastic probit and probit with flexible polynomials are nonnested. Can use Vuong's (1989, *Econometrica*) model selection test.

- If we truly believe the index structure with e heteroskedastic, there is a different way to proceed that leads to simple partial effects.
- Define the *average structural function* as a function of $\mathbf{x}$. We hold $\mathbf{x}$ at fixed values and average out the unobservable, e :

$$ASF(\mathbf{x}) = E_e\{1[\mathbf{x}\boldsymbol{\beta} + e > 0]\} = 1 - F(-\mathbf{x}\boldsymbol{\beta}),$$

where $F(\cdot)$ is the unconditional CDF of e .

- Because we do not know $F(\cdot)$, a different expression is more convenient. Let $\mathbf{x}_{i1}$ denote the random quantity. Then we can use the law of iterated expectations to show

$$ASF(\mathbf{x}) = E_{\mathbf{x}_{i1}} \{ \Phi[\exp(-\mathbf{x}_{i1}\delta)\mathbf{x}\beta] \}.$$

- A consistent estimator is

$$\widehat{ASF}(\mathbf{x}) = N^{-1} \sum_{i=1}^N \Phi[\exp(-\mathbf{x}_{i1}\hat{\delta})\mathbf{x}\hat{\beta}].$$

- The estimated average partial effect, for a continuous x_j , is

$$\widehat{APE}_j(\mathbf{x}) = \hat{\beta}_j \left\{ N^{-1} \sum_{i=1}^N \exp(-\mathbf{x}_{i1} \hat{\delta}) \phi[\exp(-\mathbf{x}_{i1} \hat{\delta}) \mathbf{x} \hat{\beta}] \right\}$$

which is the same sign as $\hat{\beta}_j$ because the term in $\{\cdot\}$ is strictly positive.

- Of course, ignoring heteroskedasticity in e does generally lead to inconsistent estimators of the β_j , but that is largely beside the point. The important question is: how far off are estimated partial effects?

- Possible point of confusion: Using the “robust” option with probit does not mean the probit estimators of β are somehow robust to heteroskedasticity in the latent error, e . In fact, β will be inconsistently estimated. [The probit MLE is still of value, providing the “best” approximation to the true response probability $p(\mathbf{x})$].
- Using “robust” means that a sandwich estimator is used for the asymptotic variance of the quasi-MLE (that is, the usual probit estimator).
- Remember, allowing heteroskedasticity in e with $y = 1[\mathbf{x}\beta + e > 0]$ changes $P(y = 1|\mathbf{x})$, which completely describes $D(y|\mathbf{x})$. This is not like other regression applications where we can have $E(y|\mathbf{x}) = \mathbf{x}\beta$ and separately talk about whether there is heteroskedasticity in $\text{Var}(y|\mathbf{x})$.